

TASK 3

Analytics:

THREADS	COARSE-GRAINED (MS)	FINE-GRAINED (MS)
2	11	70
4	23	41
8	75	79
16	235	326

Coarse-grained queues all threads on one lock, where threads are serialised but only one atomic Compare-and-Swap is in flight at a time. The lock word lives in one cache line where the CPU handles the queueing efficiently.

Fine-grained however has many locks scattered across nodes where each thread is performing Compare-and-Swap on a different node's lock word, however those lock words live on different cache lines. Then the cache-coherence protocol has to keep many cache lines coherent across 16 cores which is a lot more expensive than just one hot lock.

Therefore at 16 threads, the fine-grained locking's own overhead becomes the bottleneck not the concurrency benefit.

At 2 threads the coarse-grained list is 6 times faster than fine-grained, this is because the lock is almost never blocked whereas the fine-grained list acquires and releases a ReentrantLock on each node that it visits.

With 4 and 8 threads the coarse-grained lock becomes slightly blocked whereas the fine-grained lock continues with the same at with the 2 threads, however now it is more beneficial.

With 16 threads, the coarse-grained is surprisingly faster than the fine-grained which is not the textbook expectation. This is due to the fact that the 16 threads cause traffic across the different cache lines because the fine-grained list distributes lock words across many nodes. Coarse-grained however has one lock word which lives in one cache line.